

A triangularizing representation basis needs three vectors

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Conjecture 00000008587

The source defines adaptive bases as minimal bases making the representation matrices upper triangular and asserts that the dimension upper bound of such minimal bases is 2. We disprove this clause for a commutative operator algebra in a concrete faithful unital representation.

The operator algebra and representation

Let $H = \mathbb{C}^3$, equipped with its usual Euclidean Hilbert-space structure, and let

$$\mathcal{A} = \{aI_H : a \in \mathbb{C}\} \subseteq \text{End}_{\mathbb{C}}(H).$$

This is a unital commutative complex operator algebra. The scalar map $\rho : \mathbb{C} \rightarrow \text{End}_{\mathbb{C}}(H)$, $a \mapsto aI_H$, is an algebra homomorphism, since

$$\rho(a + b) = \rho(a) + \rho(b), \quad \rho(ab) = \rho(a)\rho(b), \quad \rho(1) = I_H.$$

It is faithful: if $aI_H = bI_H$, evaluating at the first standard basis vector gives $a = b$. Every such finite-dimensional linear operator is bounded.

For every vector-space basis B of H , the matrix of $\rho(a)$ is aI_3 . Indeed, $\rho(a)b_j = ab_j$ for each basis vector b_j . The matrix is diagonal, and therefore upper triangular. In particular the standard three-vector basis is an actual triangularizing basis for every element of $\mathcal{A}$.

Theorem 1. *The minimum cardinality of a triangularizing representation basis for $\mathcal{A}$ acting on H is 3, contradicting the asserted upper bound 2.*

Proof. The standard basis provides a triangularizing basis of cardinality 3. Every vector-space basis of H is finite and has cardinality $\dim_{\mathbb{C}} H = 3$. This applies to every possible triangularizing basis, not just the standard one. Thus the minimum is exactly 3, and no triangularizing basis has cardinality at most 2. $\square$

Meaning of minimality and scope

The bases here are vector bases of the space carrying the representation, as required to make its matrices upper triangular. Minimality is with respect to such representation bases for this fixed operator algebra acting on this fixed space. No vector can be omitted while retaining a basis of H .

We do not replace this notion by a basis of the abstract algebra $\mathcal{A}$ itself, whose algebraic dimension is 1, or minimize the dimension over different representation spaces. The source defines bases through representation matrices, and specifies no such alternative minimization. Our witness uses an actual faithful unital representation, and computes the actual representation dimension and all basis sizes. The failed upper-bound clause is a conjunct; the other claims about maximal ideal spaces and noncommutative algebras need not be resolved.

Formal objects and correspondence

The Lean space H is Mathlib's actual `EuclideanSpace  $\mathbb{C}$` (Fin 3), and `EndH` is its complex linear endomorphism algebra. The actual algebra homomorphism `scalar` is `Algebra.ofId` into this endomorphism algebra. `scalar_apply` verifies $\rho(a)x = ax$, `scalar_injective` verifies faithfulness, `scalar_unital` verifies the unit identity, and `scalar_continuous` verifies continuity of every represented scalar. `scalarAlgebra` is the actual range subalgebra. `algebra_commutative` proves that every pair of its elements commutes.

`standardBasis` is the actual standard orthonormal basis viewed as a vector-space `Basis`. `scalar_matrix` uses Mathlib's actual `LinearMap.toMatrix` and proves that the representation matrix is aI in every genuine finitely indexed basis. `Triangularizing` quantifies over every algebra element and requires each entry strictly below the diagonal to be zero. `every_basis_triangularizes` proves this property in any such ordered basis, and `standard_triangularizes` supplies its explicit standard-basis instance.

`ambient_dimension` computes the actual complex `Module.finrank` as 3. `every_basis_finite` proves that every basis index type is finite, rather than assuming this in the minimality argument. `every_basis_cardinality` proves that the actual `Nat.card` of every basis index type is 3, including arbitrary index universes. Finite-dimensionality makes this cardinal computation an ordinary finite basis count.

The theorem `minimum_triangularizing_basis_size` combines an actual triangularizing basis with the universal cardinality equality. `dimension_bound_fails` combines the actual standard basis with the impossibility of any basis having cardinality at most 2. No representation dimension, basis size or matrix data is assumed.

Reproduction and validation

The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, and `lake build`; check the entire source with

```
lake env lean -DwarningAsError=true Main.lean.
```

The source prints dependency audits for the actual representation, commutativity, matrices, triangularization, basis finiteness, cardinality and final contradiction. Only standard logical axioms are used. There are no admitted statements, custom axioms, native decision procedures or auxiliary numerical evidence.